From Unsupervised to Switching Guided Clustering

A Variational Autoencoder Implementation

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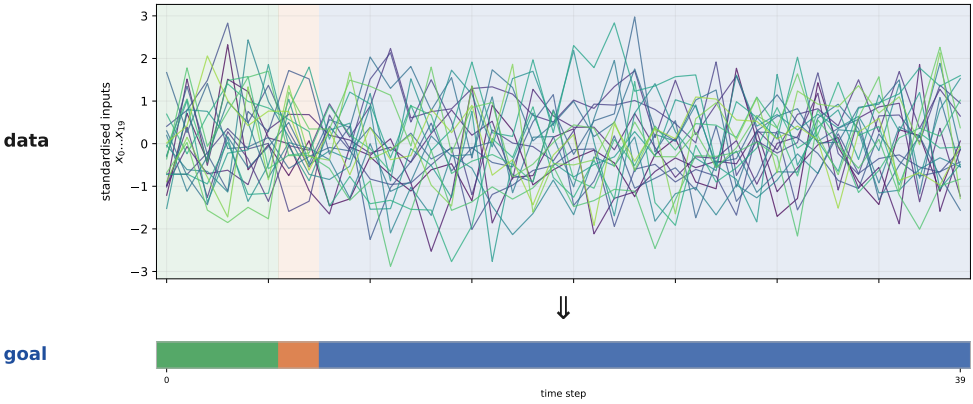
Outline

1. The goal and guided clustering principle
2. Temporal extension
3. Experiments
4. Conclusion

One individual: a multivariate time series

We observe, per individual, a multivariate sequence $\boldsymbol{x}_{1:T}$ ($T = 40$; $d_x = 20$):

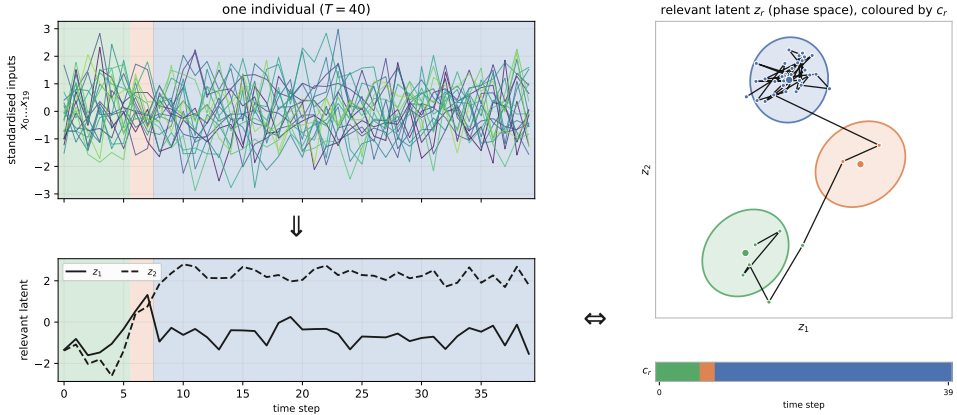
/!\ one individual, *not* 20 individuals.



Underneath runs a hidden cluster sequence $c_{1:T}$ (coloured strip) that we want to recover.

The goal: a switching clustering over time

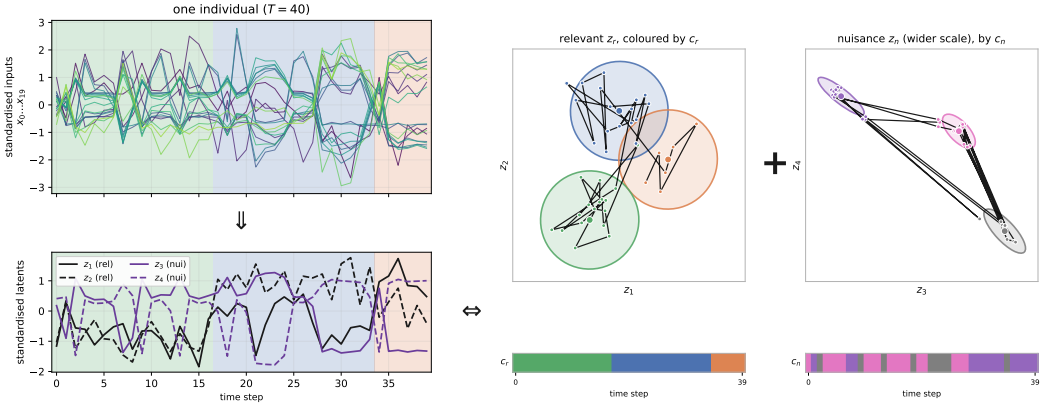
Each individual is a sequence $\boldsymbol{x}_{1:T}$ whose latent cluster switches step by step. We want to recover that switching assignment $c_{1:T}$ (the coloured strip):



A well-separated case: inputs (left), the relevant latent, and its clusters in latent space (right).

The ambiguity: several partitions co-exist

The same observations admit more than one switching partition: a subtle relevant z_r and a $\sim 15\times$ louder nuisance z_n.

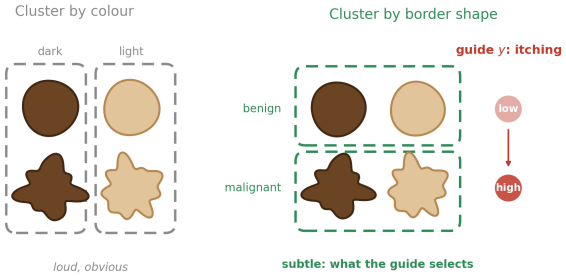


Unsupervised clustering locks onto the loud nuisance partition: how to steer to the relevant one ?

Static answer: guided clustering

A guiding variable $\boldsymbol{y}$, correlated with the relevant structure, acts as a lens that selects the task-relevant partition.

- Unlike semi-supervised: no labels on the target partition.
- Similar to predictive, but a weak $\boldsymbol{y}$ suffices.

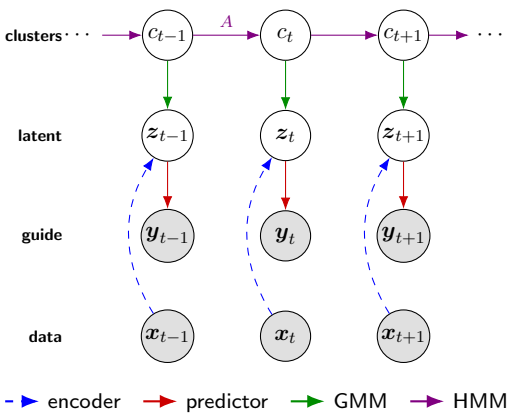


The same lesions: split by colour or by shape.

In the static case this is the Guided Clustering VAE (GCVAE). *This talk: extend it to sequences.*

GCVAE: Courrier & Biernacki, *From Unsupervised to Guided Clustering: A Variational Implementation*, arXiv:2604.05513, 2026.
Semi-supervised: Wagstaff et al., ICML 2001.
Predictive: Marbac et al., JCGS 2022; Lee & van der Schaar (AC-TPC), ICML 2020.
Gaussian-mixture latent: Jiang et al., *VaDE*, IJCAI 2017.

Hidden Markov Model - Switching Guided Clustering VAE



- **Encoder** $\boldsymbol{x}_t \dashrightarrow \boldsymbol{z}_t$: amortised $q_\phi(\boldsymbol{z}_t \mid \boldsymbol{x}_t)$.
- **Predictor** $\boldsymbol{z}_t \to \boldsymbol{y}_t$: decodes the *guide*, $p_\theta(\boldsymbol{y}_t \mid \boldsymbol{z}_t)$.
- **GMM** $c_t \to \boldsymbol{z}_t$: $p(\boldsymbol{z}_t \mid c_t{=}k) = \mathcal{N}(\boldsymbol{\mu}_k, \operatorname{diag} \boldsymbol{\sigma}_k^2)$.
- **HMM** on $c_{1:T}$: $p(c_1) = \mathrm{Cat}(\boldsymbol{\pi})$, $p(c_t \mid c_{t-1}) = \mathrm{Cat}(A_{c_{t-1},:})$.

The one new ingredient over GCVAE is the **HMM**: a Markov dependence on $c_{1:T}$ over the GMM clusters.

VAE: Kingma & Welling, *Auto-Encoding Variational Bayes*, ICLR 2014.
HMM: Rabiner, *Proc. IEEE* 1989.

Estimation loss

A β-VAE ELBO that reconstructs the guide (per sequence, normalised by T):

$$\mathcal{L} = \underbrace{\tfrac{1}{T}\,\mathbb{E}_{q_\phi(\boldsymbol{z}_{1:T}\mid\boldsymbol{x}_{1:T})}\big[\log p_\theta(\boldsymbol{y}_{1:T}\mid\boldsymbol{z}_{1:T})\big]}_{\text{reconstruct the guide } \boldsymbol{y}} - \underbrace{\tfrac{\beta}{T}\,\mathrm{KL}\big[q_\phi(\boldsymbol{z}_{1:T}\mid\boldsymbol{x}_{1:T})\,\big\|\,p_\Theta(\boldsymbol{z}_{1:T})\big]}_{\text{structure the latent via the prior } p_\Theta(\boldsymbol{z}_{1:T})}.$$

The prior marginalises over the cluster sequence $c_{1:T}$, which is where the model enters:

$$p_\Theta(\boldsymbol{z}_{1:T}) = \sum_{c_{1:T}} \underbrace{p(c_1)\prod_{t\geq 2} p(c_t\mid c_{t-1})}_{\text{HMM over clusters}} \prod_t \underbrace{\mathcal{N}\big(\boldsymbol{z}_t;\boldsymbol{\mu}_{c_t},\operatorname{diag}\boldsymbol{\sigma}^2_{c_t}\big)}_{\text{GMM emission}}.$$

K^T cluster paths, naively intractable; first-order Markov $\Rightarrow$ forward-backward, $\mathcal{O}(TK^2)$.

β-VAE: Higgins et al., ICLR 2017.
Forward-backward / Baum-Welch: Rabiner, *Proc. IEEE* 1989.

Estimation loss

Forward-backward returns the responsibilities $\gamma_t(k) = p_\Theta(c_t{=}k \mid \boldsymbol{z}_{1:T})$ and the pairwise $\xi_t(j,k) = p_\Theta(c_{t-1}{=}j, c_t{=}k \mid \boldsymbol{z}_{1:T})$. The latent KL then splits exactly into four closed-form terms $(A) + (B) + K_{\text{trans}} + K_{\text{ent}}$:

(A) encoder entropy: $-\sum_t H\big(q_\phi(\boldsymbol{z}_t \mid \boldsymbol{x}_t)\big)$

(B) emission cross-entropy: $-\mathbb{E}_{q_\phi}\big[\sum_{t,k} \gamma_t(k)\, \log p_\Theta(\boldsymbol{z}_t \mid c_t{=}k)\big]$

K_{trans} transition cross-entropy: $-\mathbb{E}_{q_\phi}\big[\sum_k \gamma_1(k) \log \pi_k + \sum_{t,j,k} \xi_t(j,k) \log A_{j,k}\big]$

K_{ent} cluster-posterior entropy: $-\mathbb{E}_{q_\phi}\big[H\big(p_\Theta(c_{1:T} \mid \boldsymbol{z}_{1:T})\big)\big]$

Variational EM: E-step = forward-backward $\Rightarrow \gamma, \xi$; M-step = closed-form ($\boldsymbol{\mu}_k, \boldsymbol{\sigma}_k$ moments; $\boldsymbol{\pi}, A$ counts). Encoder and decoder by SGD.

β-VAE: Higgins et al., ICLR 2017.
EM: Dempster, Laird & Rubin, *JRSS-B* 1977.
Forward-backward / Baum-Welch: Rabiner, *Proc. IEEE* 1989.

Implementation

- Implemented in PyTorch.
- Encoder / decoder: any network matched to the data (MLP, CNN, RNN, . . .).
- β-annealing $0 \to \beta_{\max}$ to avoid latent collapse; a low-temperature Gumbel-softmax.

Selection without labels

Bayesian optimisation on two label-free criteria. Relevance: the within-sequence guide-relevance (share of the guide's variance the clusters explain, within each sequence),

$$\eta^2_{\text{within}} = \frac{\sum_n \sum_{k=1}^{K} n_{n,k} \, (\bar{y}_{n,k} - \bar{y}_n)^2}{\sum_{n,t} (y_{n,t} - \bar{y}_n)^2}.$$

Stability: among the most relevant, prefer the configuration whose seeds agree.

Bayesian optimisation: Akiba et al., *Optuna*, KDD 2019. Gumbel-softmax: Jang, Gu & Poole, ICLR 2017.
Sticky prior: Fox et al., *HDP-HMM with state persistence*, ICML 2008.

Synthetic: recovery, and the transition matrix

ARI on the relevant cluster sequence c_r (mean $\pm$ std; 5-seed retrain of the label-free-selected model):

Generator	Unguided HMM	$T\times$GCVAE	HMM-SGCVAE
`simple`	$\mathbf{1.00}$	0.73 ± 0.02	0.78 ± 0.06
`jittered-trap`	0.00	0.33 ± 0.09	$\mathbf{0.50 \pm 0.04}$

- The unguided temporal HMM locks onto the loud nuisance (ARI $c_r{=}0.00$ on the trap): temporal structure alone is not enough, the guide is what escapes the trap.
- Wins on `jittered-trap`: the step is ambiguous but the cluster persists, so pooling over time beats the static baseline (both guided models keep ARI $c_n \leq 0.01$).
- The transition matrix is recovered faithfully (per-row correlation 1.00 on `simple`, 0.94 on `jittered-trap`): an object no static model can produce.

C-MAPSS: recovery under input noise

- **Dataset**: NASA turbofan (FD002); guide = remaining useful life. The operating condition (nuisance) is $\sim 99\%$ of the sensor variance, an unguided trap; the degradation phase (relevant) is what we want.
- **Noise**: *given the guide*, clean sensors already pin the phase at each cycle. Add per-cycle noise $\boldsymbol{x}^{\alpha}_{t,j} = \tilde{\boldsymbol{x}}_{t,j} + \alpha\,\varepsilon_{t,j}$, $\varepsilon_{t,j} \overset{\text{iid}}{\sim} \mathcal{N}(0,1)$, restores within-dwell ambiguity.

Noise level α	$T\times$GCVAE (static) ARI c_r (phase) $\uparrow$	HMM-SGCVAE ARI c_r (phase) $\uparrow$
0	0.35 ± 0.01	0.35 ± 0.01
0.1	0.20 ± 0.02	$\mathbf{0.35 \pm 0.09}$
0.2	0.09 ± 0.02	$\mathbf{0.25 \pm 0.10}$
0.5	0.03 ± 0.01	$\mathbf{0.08 \pm 0.05}$

The sticky chain pools each dwell and holds; the static per-cycle mixture chases the noise and collapses.

C-MAPSS: Saxena, Goebel, Simon & Eklund, *Damage propagation modeling for aircraft engine run-to-failure simulation*, PHM 2008.

Conclusion

- Every clustering is implicitly guided. We make that guidance explicit: a guide $\boldsymbol{y}$ as a lens onto the task-relevant partition.
- It is model-agnostic: we implement it with a VAE, and for the temporal extension we also tested an SLDS emission instead of the HMM.
- The temporal extension is safe to adopt: it never degrades the static baseline in our experiments, and improves it when a single timestep is weakly correlated with the latent dynamics (ambiguous step, persistent cluster) or under input noise.

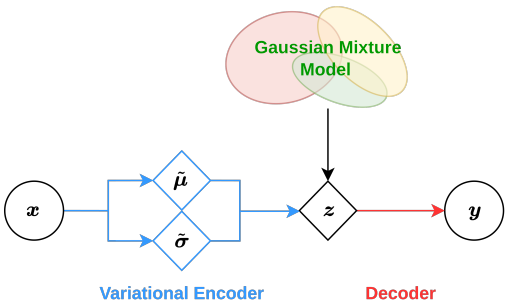
SLDS: Ghahramani & Hinton, *Variational Learning for Switching State-Space Models*, Neural Computation 2000; Linderman et al., *Recurrent Switching Linear Dynamical Systems*, AISTATS 2017.

Thank you!

Appendix: GCVAE at a glance

The Guided Clustering VAE (GCVAE), three standard building blocks:

- **Variational encoder**: projects $\boldsymbol{x}$ into a latent $\boldsymbol{z}$.
- **Gaussian mixture**: structures the latent into clusters c.
- **Predictor**: aligns clusters by predicting the guide $\boldsymbol{y}$.



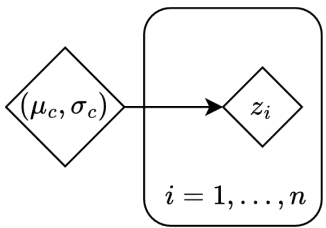
A standard VAE reconstructs $\boldsymbol{x}$; GCVAE reconstructs the guide $\boldsymbol{y}$.

Appendix: GCVAE generative model

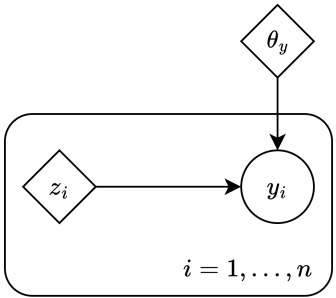
Generate the guide from a clustered latent, $c \to \boldsymbol{z} \to \boldsymbol{y}$:

$$p_\Theta(\boldsymbol{z}, c, \boldsymbol{y}) = \underbrace{\mathrm{Cat}(c; \boldsymbol{\pi})}_{p(c)} \underbrace{\mathcal{N}(\boldsymbol{z}; \boldsymbol{\mu}_c, \boldsymbol{\sigma}_c^2 I)}_{p_\Theta(\boldsymbol{z}|c)} \underbrace{\mathcal{N}(\boldsymbol{y}; f_\theta(\boldsymbol{z}), I)}_{p_\theta(\boldsymbol{y}|\boldsymbol{z})}$$

GMM: choose c, draw $\boldsymbol{z} \mid c$



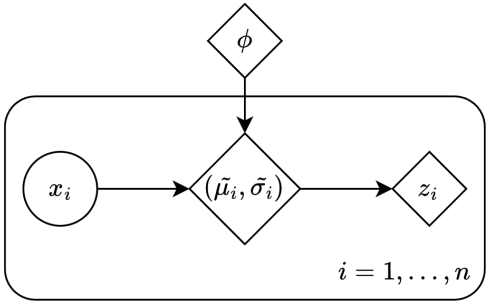
Predictor: draw $\boldsymbol{y} \mid \boldsymbol{z}$



Appendix: GCVAE inference model

The posterior $p_\Theta(\boldsymbol{z}, c \mid \boldsymbol{y})$ is intractable; a variational encoder infers the latent from $\boldsymbol{x}$ (mean-field):

$$q_\phi(\boldsymbol{z}, c \mid \boldsymbol{x}) = q_\phi(\boldsymbol{z} \mid \boldsymbol{x})\, q(c \mid \boldsymbol{x}), \qquad q_\phi(\boldsymbol{z} \mid \boldsymbol{x}) = \mathcal{N}(\boldsymbol{z}; \tilde{\boldsymbol{\mu}}, \tilde{\boldsymbol{\sigma}}^2 I), \quad [\tilde{\boldsymbol{\mu}}, \log \tilde{\boldsymbol{\sigma}}^2] = g_\phi(\boldsymbol{x})$$



Variational encoder: $\boldsymbol{x} \to (\tilde{\boldsymbol{\mu}}, \tilde{\boldsymbol{\sigma}}) \to \boldsymbol{z}$, giving $q_\phi(\boldsymbol{z} \mid \boldsymbol{x})$.

Appendix: GCVAE ELBO

The posterior $p_\Theta(\boldsymbol{z}, c \mid \boldsymbol{y})$ is intractable, so we fit $q_\phi(\boldsymbol{z}, c \mid \boldsymbol{x})$ (encode $\boldsymbol{x}$, target $\boldsymbol{y}$) by

$$\min_{\phi} \; \mathrm{KL}\big[\, q_\phi(\boldsymbol{z}, c \mid \boldsymbol{x}) \;\big\|\; p_\Theta(\boldsymbol{z}, c \mid \boldsymbol{y}) \,\big] .$$

Using $p_\Theta(\boldsymbol{z}, c, \boldsymbol{y}) = p_\theta(\boldsymbol{y} \mid \boldsymbol{z})\, p_\Theta(\boldsymbol{z}, c)$:

$$\begin{aligned} \mathrm{KL}\big[q_\phi(\boldsymbol{z}, c \mid \boldsymbol{x}) \,\big\|\, p_\Theta(\boldsymbol{z}, c \mid \boldsymbol{y})\big] &= \log p_\Theta(\boldsymbol{y}) - \mathbb{E}_{q_\phi(\boldsymbol{z}, c \mid \boldsymbol{x})}[\log p_\theta(\boldsymbol{y} \mid \boldsymbol{z})] \\ &\quad + \mathrm{KL}\big[q_\phi(\boldsymbol{z}, c \mid \boldsymbol{x}) \,\big\|\, p_\Theta(\boldsymbol{z}, c)\big] . \end{aligned}$$

The left-hand side is ≥ 0 and $\log p_\Theta(\boldsymbol{y})$ is constant in ϕ, so $\log p_\Theta(\boldsymbol{y}) \geq \mathrm{ELBO}$ with

$$\mathrm{ELBO} = \underbrace{\mathbb{E}_{q_\phi(\boldsymbol{z}, c \mid \boldsymbol{x})}[\log p_\theta(\boldsymbol{y} \mid \boldsymbol{z})]}_{\text{reconstruct the guide}} - \underbrace{\mathrm{KL}\big[q_\phi(\boldsymbol{z}, c \mid \boldsymbol{x}) \,\big\|\, p_\Theta(\boldsymbol{z}, c)\big]}_{\text{structure the latent (GMM)}} .$$

β-VAE: maximise the ELBO with the KL term weighted by β. The switching model extends exactly this to $T > 1$.

β-VAE: Higgins et al., ICLR 2017.
GCVAE: Courrier & Biernacki, arXiv:2604.05513, 2026.

Appendix: Withings, per-night guide relevance

Per-night guide relevance $\eta^2(c \to \mathrm{AHI})$ on a 14-night connected-health cohort, static baseline against the dynamic model:

K	$T\times$GCVAE (static)	HMM-SGCVAE (dynamic)
3	0.17	0.21
9	0.20	0.25
14	0.21	0.25

The dynamic model edges the static baseline at every cluster count K.